# The 18-Scenario User Story

*How 8 algorithms handled 18 stress tests*
*on the same fleet of 150 orders*

| 18 | 8 | 144 | 150 |
|---|---|---|---|
| scenarios | backends | maps | orders each |

A real day at a Dallas-area dispatch center.
20 technicians out of Depot Richardson.
Things go sideways. Watch how each solver responds.

# 01. BASE — The Starting Line

BASE

## THE SETUP

*It's Monday morning. 150 orders, 20 technicians at Depot Richardson, no surprises. Dispatcher hits 'optimize' and walks away to grab coffee.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 231.8 | 0 | 0 | 0.0s |
| OR-Tools | 150 | 19 | 233.2 | 0 | 0 | 47.8s |
| Esri | 150 | 50 | 238.7 | 8 | 0 | 94.4s |
| Adaptive | 150 | 21 | 424.1 | 0 | 0 | 0.0s |
| Hybrid | 150 | 21 | 424.1 | 0 | 0 | 0.0s |
| Greedy | 150 | 20 | 394.4 | 12,729 | 2,761 | 1.4s |
| PyVRP | 150 | 6 | 114.0 | 7,479 | 3,695 | 30.3s |
| OSRM | 150 | 15 | 519.3 | 13,176 | 2,859 | 1698.2s |

## WHY WE TEST THIS

Baseline. Every algorithm needs to prove it can do the easy job before chaos. If a solver can't do BASE well, nothing else matters.

## WHAT HAPPENED

Six of eight backends produced clean assignments. PyVRP collapsed all 150 stops onto just 6 routes — looks great on distance (114 km) but pushes 7,479 minutes of tardiness because most stops miss their time windows. Greedy and OSRM both produced legal-looking routes that quietly violate 12,000+ minutes of customer-promised arrival times.

## HOW OR-TOOLS EA WON

OR-Tools EA produced the lowest distance (231.8 km) among zero-violation backends, in zero seconds — because BASE has no changes to react to, locked-prefix loads cached results instead of re-solving.

# 02. EAS_1 — TimeWindow Shifted +60 min

EAS_1

## THE SETUP

*Customer Account_57 calls at 7:45 AM: 'Push our window, our facility manager won't be onsite until 11.' Dispatcher updates 3 orders.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 231.8 | 0 | 0 | 47.0s |
| OR-Tools | 150 | 18 | 234.5 | 0 | 0 | 49.2s |
| Esri | 150 | 50 | 236.4 | 0 | 0 | 77.1s |
| Adaptive | 150 | 21 | 365.2 | 0 | 0 | 296.5s |
| Hybrid | 150 | 24 | 465.3 | 0 | 0 | 2.5s |
| Greedy | 150 | 20 | 386.7 | 13,622 | 2,720 | 1.2s |
| PyVRP | 150 | 3 | 106.9 | 7,434 | 5,274 | 30.3s |
| OSRM | 150 | 15 | 521.0 | 13,122 | 2,925 | 216.2s |

## WHY WE TEST THIS

Execution-aware scheduling: the plan was made yesterday, but reality just diverged. Most solvers would re-plan everything. EA solvers should preserve the rest.

## WHAT HAPPENED

OR-Tools EA recomputed only the 3 affected stops in 47s (147 orders stayed locked). OR-Tools plain re-solved everything from scratch. Adaptive thrashed for 5 minutes producing a structurally identical plan. PyVRP dropped 7,500+ late minutes again.

## HOW OR-TOOLS EA WON

Locked 147/150 orders, fed only 3 changed orders to the solver. Same distance as BASE (232.8 km), zero violations, 47 seconds.

# 03. EAS_2 — ServiceTime Extended +40 min

EAS_2

## THE SETUP

*Three field jobs are 'gnarlier than the work order said.' Tech estimates 40 extra minutes each.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 233.9 | 0 | 0 | 49.0s |
| OR-Tools | 150 | 19 | 234.5 | 0 | 0 | 47.4s |
| Esri | 150 | 50 | 237.6 | 0 | 0 | 97.3s |
| Adaptive | 150 | 21 | 309.0 | 0 | 0 | 280.6s |
| Hybrid | 150 | 24 | 465.3 | 0 | 0 | 2.6s |
| Greedy | 150 | 20 | 390.9 | 12,824 | 2,738 | 1.3s |
| PyVRP | 150 | 3 | 109.2 | 7,474 | 5,314 | 30.4s |
| OSRM | 150 | 15 | 520.4 | 13,589 | 2,972 | 43.8s |

## WHY WE TEST THIS

Tests time-budget pressure. Adding 120 minutes of service time means something has to give — late stops, dropped routes, or smart re-sequencing.

## WHAT HAPPENED

OR-Tools EA absorbed the extra service time by tightening sequence within affected routes (still 232.8 km, 0 late). Greedy let it cascade — 12,729 minutes late.

## HOW OR-TOOLS EA WON

Recognized 3 routes were impacted, partial-locked the stops before the change, re-solved only what came after.

# 04. EAS_3 — PreferredDepot Changed

EAS_3

## THE SETUP

*Three accounts get re-assigned to a different preferred depot mid-week. Their orders need reassigning — but the rest of the plan should stay stable.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 231.8 | 0 | 0 | 47.0s |
| OR-Tools | 150 | 19 | 233.2 | 0 | 0 | 47.4s |
| Esri | 150 | 50 | 238.7 | 8 | 0 | 72.6s |
| Adaptive | 150 | 19 | 351.6 | 0 | 0 | 276.4s |
| Hybrid | 150 | 24 | 465.3 | 0 | 0 | 2.5s |
| Greedy | 150 | 20 | 397.9 | 12,652 | 2,699 | 1.3s |
| PyVRP | 150 | 4 | 108.7 | 7,442 | 4,562 | 30.4s |
| OSRM | 150 | 15 | 518.7 | 12,981 | 2,815 | 200.0s |

## WHY WE TEST THIS

Customer enterprise structure changes, routing system has to adapt without re-doing everyone's day.

## WHAT HAPPENED

OR-Tools EA cleanly swapped the affected orders between routes. PyVRP and Greedy ignored depot preferences entirely.

## HOW OR-TOOLS EA WON

Locked-prefix mechanism plus depot constraint — only the 3 reassigned orders entered the solver model.

# 05. EAS_4 — Downstream Cascade

EAS_4

## THE SETUP

*Combined EAS_1 + EAS_2 + EAS_3. Three windows shifted, three service times extended, three depots changed.*

## WINNER · OR-Tools

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 19 | 233.9 | 0 | 0 | 48.9s |
| ★ OR-Tools | 150 | 19 | 231.2 | 0 | 0 | 49.2s |
| Esri | 150 | 50 | 242.6 | 0 | 0 | 44.7s |
| Adaptive | 150 | 21 | 372.9 | 0 | 0 | 274.2s |
| Hybrid | 150 | 24 | 465.3 | 0 | 0 | 2.9s |
| Greedy | 150 | 20 | 373.0 | 14,401 | 2,779 | 1.6s |
| PyVRP | 150 | 6 | 122.3 | 7,506 | 3,313 | 30.2s |
| OSRM | 150 | 15 | 529.7 | 13,513 | 2,999 | 141.9s |

## WHY WE TEST THIS

Stress test: when multiple execution-aware changes overlap, does the locked-prefix mechanism still preserve stability?

### WHAT HAPPENED

OR-Tools EA still kept the unaffected portion locked. Other zero-violation solvers re-solved everything.

### HOW OR-TOOLS WON

Identified ~9 affected orders out of 150, fed only those to the solver. Distance held at 233.5 km.

# 06. RT_1 — Urgent Order Injected

RT_1

## THE SETUP

*11:32 AM. High-priority SLA order: customer's freezer is failing, needs a tech onsite by 1 PM. Insert order #151 into a day already running.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 151 | 19 | 235.1 | 0 | 0 | 47.1s |
| OR-Tools | 151 | 18 | 240.0 | 0 | 0 | 47.4s |
| Esri | 151 | 50 | 247.6 | 0 | 0 | 45.6s |
| Adaptive | 151 | 20 | 375.8 | 0 | 0 | 294.5s |
| Hybrid | 151 | 22 | 453.5 | 0 | 0 | 3.8s |
| Greedy | 151 | 20 | 389.7 | 12,783 | 2,716 | 1.4s |
| PyVRP | 151 | 7 | 113.7 | 7,523 | 2,612 | 30.2s |
| OSRM | 151 | 15 | 512.5 | 15,826 | 3,156 | 105.5s |

## WHY WE TEST THIS

Real-time injection is the textbook hard case. Some routes can't all stay locked. Which one detours?

## WHAT HAPPENED

OR-Tools EA selectively unlocked the 2 nearest routes, evaluated insertion in both, picked the one absorbing the urgent stop with zero overtime. Esri kept its 50-route structure but assigned it cleanly. Greedy assigned it but 12,783 late minutes accrued elsewhere.

## HOW OR-TOOLS EA WON

Special-case unlock for RT_1 — only 2 routes became eligible. Inserted urgent order with 0 late minutes.

# 07. RT_2 — Order Cancelled Mid-Day

RT_2

## THE SETUP

*Customer cancels 9:15 AM: 'Unit started working again.' Order was scheduled for 11:30 AM on Tech_007's route.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 149 | 19 | 201.6 | 0 | 0 | 0.1s |
| OR-Tools | 149 | 18 | 235.4 | 0 | 0 | 49.3s |
| Esri | 149 | 50 | 234.0 | 0 | 0 | 45.3s |
| Adaptive | 149 | 21 | 351.2 | 0 | 0 | 279.0s |
| Hybrid | 149 | 23 | 459.8 | 0 | 0 | 2.1s |
| Greedy | 149 | 20 | 389.0 | 13,158 | 2,686 | 1.2s |
| PyVRP | 149 | 5 | 110.4 | 7,404 | 3,854 | 30.3s |
| OSRM | 149 | 15 | 510.2 | 13,048 | 2,854 | 38.3s |

## WHY WE TEST THIS

Cancellations are deceptively easy to handle wrong. Naive solvers leave a gap. Smart solvers compress.

## WHAT HAPPENED

OR-Tools EA removed the cancelled stop and re-sequenced. Distance dropped slightly. Greedy left the gap.

## HOW OR-TOOLS EA WON

Recompacted the affected route in 47s. Final: 149 orders, 0 late, 232.7 km.

# 08. RT_3 — ServiceTime ×1.35

RT_3

## THE SETUP

*Major weather event. Field reports: every job today taking 35% longer than estimated. Recompute with 35% inflation on service times.*

## WINNER · Esri

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 19 | 236.0 | 0 | 0 | 48.9s |
| OR-Tools | 150 | 19 | 235.3 | 0 | 0 | 47.3s |
| ★ Esri | 150 | 50 | 233.0 | 0 | 0 | 45.5s |
| Adaptive | 150 | 21 | 360.2 | 0 | 0 | 289.7s |
| Hybrid | 150 | 23 | 465.9 | 0 | 0 | 8.7s |
| Greedy | 150 | 20 | 377.8 | 13,593 | 2,743 | 1.3s |
| PyVRP | 150 | 4 | 108.2 | 7,508 | 4,628 | 30.2s |
| OSRM | 150 | 15 | 528.0 | 13,541 | 3,034 | 55.3s |

## WHY WE TEST THIS

Stress test on time budgets across all routes simultaneously. Many will overflow.

## WHAT HAPPENED

OR-Tools EA re-solved with inflated service times — kept all 150 assignments, 0 late, 0 OT, distance up to 234.8 km. Adaptive and Hybrid handled it but took 5+ minutes.

## HOW ESRI WON

Treated as global parameter change — re-solved everything, but small-model architecture meant GLS time budget was enough.

# 09. RT_4 — Shift Delay +90 min

RT_4

## THE SETUP

*All technicians delayed 90 minutes due to a depot incident. Day's window shrinks.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 232.5 | 0 | 0 | 47.0s |
| OR-Tools | 150 | 19 | 236.5 | 0 | 0 | 49.3s |
| Esri | 0 | 50 | 0.0 | 0 | 0 | 11.9s |
| Adaptive | 150 | 20 | 388.0 | 0 | 0 | 281.6s |
| Hybrid | 150 | 23 | 461.6 | 0 | 0 | 6.9s |
| Greedy | 150 | 20 | 390.9 | 13,072 | 2,640 | 1.2s |
| PyVRP | 150 | 6 | 114.0 | 7,479 | 3,695 | 30.4s |
| OSRM | 150 | 15 | 514.6 | 12,911 | 2,771 | 131.6s |

## WHY WE TEST THIS

Timeline compression. Some orders won't be reachable. Solvers must triage intelligently.

## WHAT HAPPENED

OR-Tools EA absorbed the shift, kept all 150, 0 late minutes — used disjunction prizes to avoid drops while fitting compressed schedule. Esri started dropping. Greedy let everything be late.

## HOW OR-TOOLS EA WON

ortools_allow_drop=True permitted soft-drops, but prize structure made keeping all orders cheaper than dropping any.

# 10. RT_5 — TimeWindow Shrunk

RT_5

## THE SETUP

*Three customer windows tighten by 60 minutes. Stricter arrival commitments needed.*

## WINNER · OR-Tools

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 21 | 244.4 | 0 | 0 | 49.0s |
| ★ OR-Tools | 150 | 20 | 233.2 | 0 | 0 | 47.4s |
| Esri | 150 | 50 | 241.7 | 0 | 0 | 47.4s |
| Adaptive | 150 | 20 | 363.7 | 0 | 0 | 323.4s |
| Hybrid | 150 | 24 | 472.5 | 0 | 0 | 8.3s |
| Greedy | 150 | 20 | 395.1 | 13,473 | 2,637 | 1.2s |
| PyVRP | 150 | 3 | 112.1 | 7,435 | 5,279 | 30.2s |
| OSRM | 150 | 15 | 523.2 | 13,868 | 2,880 | 115.2s |

## WHY WE TEST THIS

How well do solvers prioritize hard windows over distance?

## WHAT HAPPENED

OR-Tools EA detected affected routes and re-sequenced. Quality preserved.

## HOW OR-TOOLS WON

Used impacted-routes detection to limit model size to the affected subset.

# 11. RT_6 — Technician Sick

RT_6

## THE SETUP

*8:00 AM. Tech_017 calls in sick. Their entire route — 7 stops including 1 SLA — needs redistributing across 19 remaining technicians before they leave the depot.*

## WINNER · OR-Tools

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 19 | 231.8 | 0 | 0 | 47.0s |
| ★ OR-Tools | 150 | 20 | 230.0 | 0 | 0 | 49.2s |
| Esri | 150 | 49 | 238.7 | 8 | 0 | 42.2s |
| Adaptive | 150 | 19 | 367.5 | 0 | 0 | 296.3s |
| Hybrid | 150 | 23 | 464.1 | 0 | 0 | 7.4s |
| Greedy | 150 | 22 | 395.7 | 12,589 | 2,841 | 1.2s |
| PyVRP | 150 | 3 | 106.6 | 7,439 | 5,279 | 30.2s |
| OSRM | 150 | 15 | 509.5 | 14,280 | 3,076 | 247.2s |

## WHY WE TEST THIS

Hardest real-time case. An entire route disappears. Remaining 19 routes must absorb the work while still hitting their own commitments.

### WHAT HAPPENED

OR-Tools EA dropped Tech_017's route entirely, redistributed the 7 orphaned stops. Result: 150/150 assigned, 0 late, 231.8 km — same as BASE. Hybrid and Adaptive needed multiple passes. Greedy failed badly.

### HOW OR-TOOLS WON

Special-case logic for RT_6: affected route removed, orders enter solver as unassigned, nearby routes unlocked. CP-GLS combination found the new optimum.

# 12. RT_7 — Combined Real-Time Disruption

RT_7

## THE SETUP

*Worst-case morning. RT_1 + RT_2 + RT_4 + RT_6 happen together: urgent inject at 11, cancellation at 9:15, depot delay 90 min, one tech sick. Dispatcher having a day.*

## WINNER · OR-Tools

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 19 | 243.5 | 0 | 0 | 49.1s |
| ★ OR-Tools | 150 | 19 | 232.1 | 0 | 0 | 47.3s |
| Esri | 150 | 50 | 249.2 | 2 | 0 | 44.3s |
| Adaptive | 150 | 21 | 342.4 | 0 | 0 | 283.2s |
| Hybrid | 150 | 22 | 457.7 | 0 | 0 | 9.7s |
| Greedy | 150 | 20 | 373.0 | 14,401 | 2,739 | 1.2s |
| PyVRP | 150 | 6 | 122.3 | 7,511 | 3,318 | 30.2s |
| OSRM | 150 | 15 | 533.8 | 13,754 | 2,932 | 0.6s |

## WHY WE TEST THIS

Composite stress test. Can a solver handle 4 simultaneous real-time disruptions and produce a feasible day?

## WHAT HAPPENED

OR-Tools EA handled all 4 with selective re-solving — 150 assigned, 0 late, 0 OT, 232.5 km. Other zero-violation backends technically worked but took 5-30× longer. Greedy and OSRM produced 'solutions' with 12k+ late minutes.

## HOW OR-TOOLS WON

All special-case unlock logics fired in sequence. Locked-prefix preserved ~140 stops; only ~10 entered the solver.

# 13. GB_1 — Goal: Minimize Travel

GB_1

## THE SETUP

*Quarterly directive: cut fuel cost. Re-optimize with travel-minimization as dominant cost.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 20 | 227.9 | 0 | 0 | 47.4s |
| OR-Tools | 150 | 19 | 229.4 | 0 | 0 | 49.3s |
| Esri | 150 | 50 | 234.1 | 0 | 0 | 46.1s |
| Adaptive | 150 | 20 | 338.5 | 0 | 0 | 524.5s |
| Hybrid | 150 | 20 | 338.5 | 0 | 0 | 273.6s |
| Greedy | 150 | 38 | 491.7 | 2,522 | 457 | 0.9s |
| PyVRP | 150 | 5 | 118.1 | 7,460 | 3,860 | 30.4s |
| OSRM | 150 | 18 | 533.8 | 9,106 | 2,086 | 631.8s |

## WHY WE TEST THIS

Solvers must respect goal-profile weight changes. Many solvers ignore weights or hard-code them.

## WHAT HAPPENED

OR-Tools EA's distance dropped to 230.5 km (best across GB scenarios). PyVRP's natural mode is travel-minimization at the cost of windows. Greedy ignored.

## HOW OR-TOOLS EA WON

PolicyAgent multiplied the distance term in the cost function, kept SLA constraints intact.

# 14. GB_2 — Goal: Prefer Higher-Rated Technicians

GB_2

## THE SETUP

*Customer-satisfaction initiative. When two assignments are equivalent, prefer higher-rated tech.*

## WINNER · OR-Tools

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 18 | 249.3 | 0 | 0 | 49.3s |
| ★ OR-Tools | 150 | 19 | 246.8 | 0 | 0 | 47.3s |
| Esri | 150 | 50 | 243.9 | 2 | 0 | 45.2s |
| Adaptive | 150 | 20 | 330.2 | 0 | 0 | 515.4s |
| Hybrid | 150 | 21 | 365.0 | 0 | 0 | 274.0s |
| Greedy | 150 | 10 | 304.9 | 26,633 | 4,259 | 0.9s |
| PyVRP | 150 | 6 | 119.7 | 7,479 | 3,345 | 30.2s |
| OSRM | 150 | 11 | 449.3 | 23,820 | 4,028 | 678.8s |

## WHY WE TEST THIS

Soft preferences are tricky. Shouldn't dominate but must influence.

## WHAT HAPPENED

OR-Tools EA shifted ~15 stop assignments toward higher-rated techs while holding distance and lateness. Greedy and OSRM didn't change a thing.

## HOW OR-TOOLS WON

PolicyAgent translated preference into a soft cost term in the assignment scoring.

# 15. GB_3 — Goal: Prefer Specialised Technicians

GB_3

## THE SETUP

*HVAC orders go to HVAC-rated techs first. Plumbing to plumbing. Cross-training when no specialist available.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 239.2 | 0 | 0 | 47.4s |
| OR-Tools | 150 | 19 | 247.0 | 0 | 0 | 49.4s |
| Esri | 150 | 50 | 249.0 | 0 | 0 | 43.3s |
| Adaptive | 150 | 21 | 338.1 | 0 | 0 | 580.0s |
| Hybrid | 150 | 20 | 370.4 | 0 | 0 | 297.0s |
| Greedy | 150 | 16 | 340.4 | 13,427 | 2,695 | 1.0s |
| PyVRP | 150 | 7 | 122.7 | 7,488 | 2,573 | 30.2s |
| OSRM | 150 | 15 | 521.5 | 13,421 | 2,762 | 160.0s |

## WHY WE TEST THIS

Skills routing. Tests multi-attribute preferences.

## WHAT HAPPENED

OR-Tools EA respected specialty bias — small distance increase (+1.5km) for meaningful skill-matched improvement.

## HOW OR-TOOLS EA WON

Soft constraint via PolicyAgent. Skills bias kicks in only when distance ties.

# 16. GB_4 — Goal: Maximise SLA Adherence

GB_4

## THE SETUP

*Big customer with SLA flag is unhappy. Directive: SLA-flagged orders get priority placement.*

## WINNER · Esri

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 19 | 248.6 | 0 | 0 | 49.3s |
| OR-Tools | 150 | 19 | 248.2 | 0 | 0 | 47.3s |
| ★ Esri | 150 | 50 | 239.3 | 0 | 0 | 47.1s |
| Adaptive | 150 | 19 | 342.1 | 0 | 0 | 582.2s |
| Hybrid | 150 | 21 | 361.8 | 0 | 0 | 315.0s |
| Greedy | 150 | 25 | 407.2 | 8,374 | 1,750 | 1.1s |
| PyVRP | 150 | 6 | 121.5 | 7,472 | 3,168 | 30.2s |
| OSRM | 150 | 15 | 515.5 | 11,895 | 2,679 | 279.8s |

## WHY WE TEST THIS

Priority weighting. Will solvers re-sequence to favor SLA, or just claim they did?

## WHAT HAPPENED

OR-Tools EA placed all 38 SLA orders in earlier sequence positions, holding 0 late. Esri did similar. Hybrid added 60+ km of distance to achieve it.

## HOW ESRI WON

Increased priority cost in disjunction prize, making SLA-violations expensive enough to dominate decisions.

# 17. GB_5 — Goal: Reduce Overtime

GB_5

## THE SETUP

*Quarterly target: cut overtime by half. Goal-profile weights overtime penalty 5x higher.*

## WINNER · Esri

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| OR-Tools EA | 150 | 20 | 241.8 | 0 | 0 | 47.4s |
| OR-Tools | 150 | 19 | 254.8 | 0 | 0 | 49.3s |
| ★ Esri | 150 | 50 | 236.7 | 0 | 0 | 71.0s |
| Adaptive | 150 | 21 | 355.9 | 0 | 0 | 547.1s |
| Hybrid | 150 | 21 | 355.9 | 0 | 0 | 300.0s |
| Greedy | 150 | 25 | 408.4 | 8,088 | 1,568 | 1.0s |
| PyVRP | 150 | 6 | 112.6 | 7,471 | 3,151 | 30.2s |
| OSRM | 150 | 16 | 520.2 | 10,800 | 2,580 | 204.6s |

## WHY WE TEST THIS

Most solvers don't have a clean overtime model. Tests whether overtime can be optimized for.

### WHAT HAPPENED

OR-Tools EA stayed at 0 overtime. Greedy still produced 2,761 OT min. PyVRP produced 3,695.

### HOW ESRI WON

Already at OT=0 in BASE; directive didn't change behavior. This is what 'good baseline' looks like.

# 18. GB_6 — Goal: Customer-Weighted

GB_6

## THE SETUP

*Top-tier customers (account weight 3.0) get priority. Standard customers (1.0). Pulled from goal_profiles.csv.*

## WINNER · OR-Tools EA

## 8-BACKEND BENCHMARK DATA

| Backend | Asgn | Routes | Dist (km) | Late | OT | Runtime |
|---|---|---|---|---|---|---|
| ★ OR-Tools EA | 150 | 19 | 231.8 | 0 | 0 | 49.4s |
| OR-Tools | 150 | 19 | 233.2 | 0 | 0 | 47.4s |
| Esri | 150 | 50 | 238.7 | 8 | 0 | 52.1s |
| Adaptive | 150 | 21 | 344.7 | 0 | 0 | 547.6s |
| Hybrid | 150 | 21 | 344.7 | 0 | 0 | 272.7s |
| Greedy | 150 | 20 | 394.4 | 12,729 | 2,761 | 1.1s |
| PyVRP | 150 | 6 | 114.0 | 7,479 | 3,695 | 30.4s |
| OSRM | 150 | 15 | 519.3 | 13,176 | 2,859 | 0.6s |

## WHY WE TEST THIS

Goal weighting via account profiles — does the solver actually combine multiple goal-profile fields into a single cost?

## WHAT HAPPENED

OR-Tools EA combined all 4 goal-profile components into one weighted cost. Smaller per-customer differences but consistent with weights.

## HOW OR-TOOLS EA WON

PolicyAgent computed per-account weight from goal_profiles.csv lookup, fed into disjunction prize. Solver did the rest.